SERVER Project



10281 - Managing Open Adapter Card (OAC)

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OCP PLATINUM Member

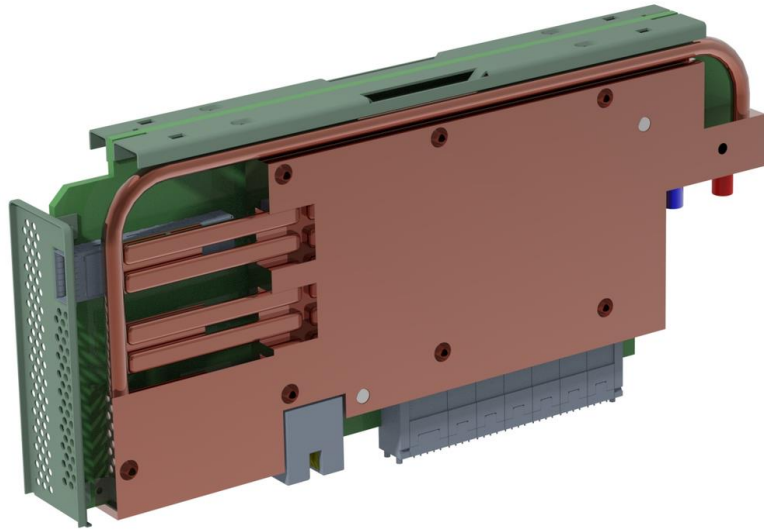
What You'll Leave With

1. **Why OAC management looks the way it does** Designed from a blank slate for AI infrastructure — not inherited from CEM or OAM.

2. **How an OAC is managed inside a system** One always-available managed endpoint per adapter, aggregated up to the BMC.

3. **The two innovations that make it work** CID replaces mechanical keying. SCD replaces strap pins — kilobytes where there used to be bits.

4. **Why the OCP alignment matters to you** Management Interfaces, RAS, and Firmware Update all apply over that same path.



The OAC Design Philosophy

START FROM A BLANK SLATE

Nothing inherited.

OAC management did not begin from an existing card's conventions.

Every mechanism had to earn its place, and the legacy adapter-management practices accumulated across PCIe CEM and OAM was deliberately left behind.

A MODERN WAY TO MANAGE A CARD

Declare, don't strap.

Identity, capability, and slot configuration are management objects exchanged on the management interface, **not** resistor straps and mechanical keys.

Management Software establishes compatibility before anything is powered on.

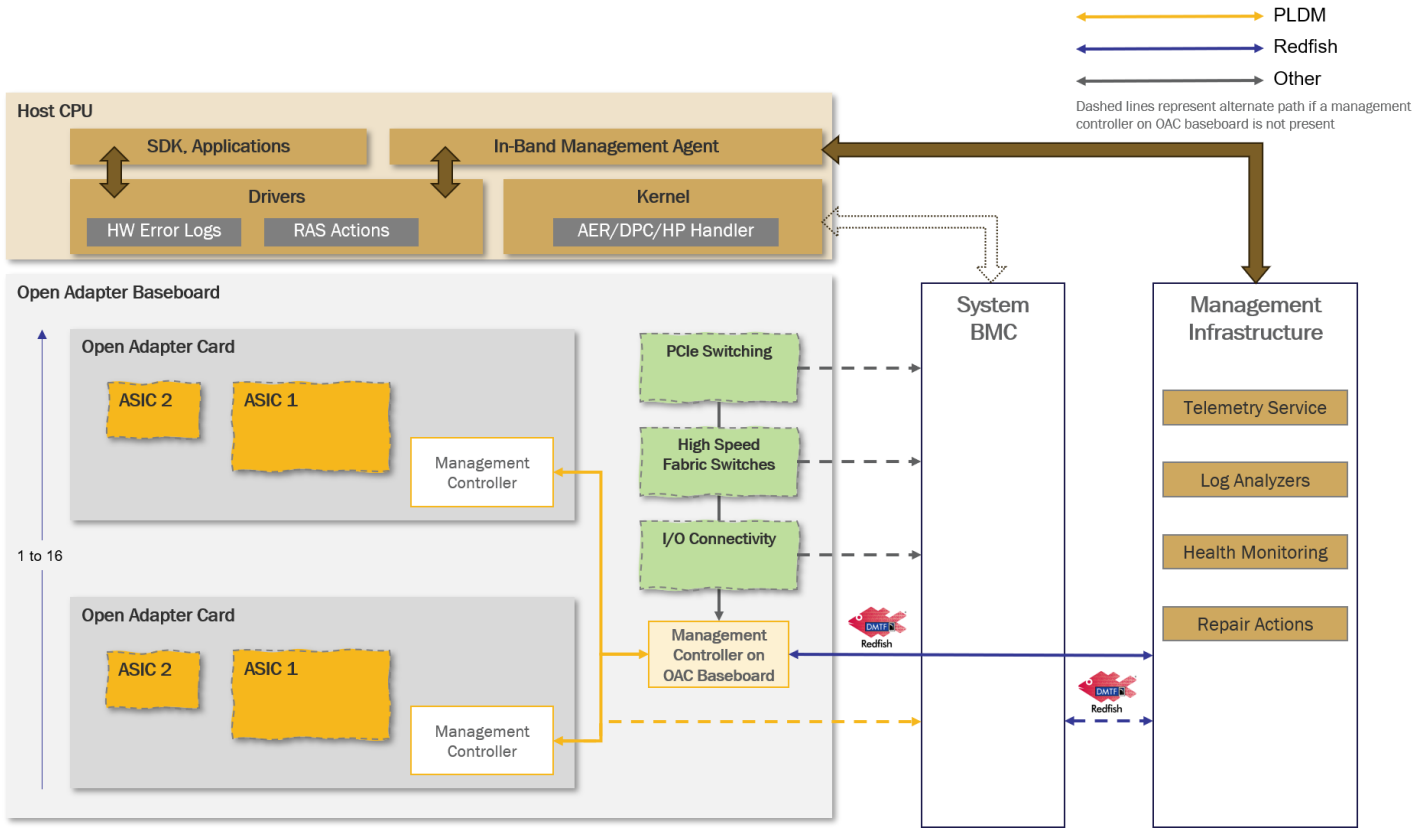
BUILT FOR AI INFRASTRUCTURE

Designing for the future.

Dense accelerator racks, mixed protocols sharing one mechanical envelope, fleet-scale telemetry, and adapters whose power draw has to be understood before they are switched on.

The result: One mandatory management path, and configuration expressed as data.

How an OAC is managed inside a system



Example Management System Level View

The topology above is illustrative. The Managed Accelerator Subsystem aggregates adapters and abstracts the accelerator infrastructure from the rest of the system. The Base Specification fixes the interface and the protocol to the OAC — not the hierarchy above it.

USB first. Management rides USB 2.0 on CONN0. One transport for discovery, telemetry, RAS and firmware update, on every compliant adapter.

An endpoint that is always available. Each OAC carries a management controller, alive on +12 V auxiliary power (12W).

Industry standard protocols. PLDM over MCTP over USB at the OAC boundary, Redfish above it.

Single Mandatory Management Interface

Every compliant OAC exposes MCTP over USB 2.0 with PLDM on CONN0

OAC_USB2_MC_D+ / D− — Column 11, statically assigned, always present, available on +12 V auxiliary power alone

CONN0 COL 11		CONN0 COL 10	
A11	OAC_PWRBRK#	A10	GND
B11	GND	B10	OAC_SB_RFU4
C11	OAC_PWR_GD	C10	OAC_SB_RFU3
D11	OAC_PWR_EN	D10	GND
E11	GND	E10	OAC_SB_RFU2
F11	OAC_MC_RST#	F10	OAC_SB_RFU1
G11	OAC_MC_RDY	G10	GND
H11	GND	H10	OAC_SB_RFU0
I11	**OAC_USB2_MC_D−**	I10	OAC_PLT_RST#
J11	**OAC_USB2_MC_D+**	J10	GND
K11	GND	K10	OAC_USB2_DBG_D−
L11	+12V AUX	L10	OAC_USB2_DBG_D+
M11	+12V AUX	M10	GND
N11	GND	N10	OAC_AUX_PWR_GD

CARRIED ON THIS ONE PATH

- Adapter discovery and MCTP endpoint enumeration
- FRU and inventory access
- CID retrieval
- SCD configuration
- Telemetry, health and RAS reporting
- Firmware update
- Interoperability evaluation and power-enable coordination

EXPLICITLY NOT REQUIRED

- Dynamically assigned pins, cabled interfaces, vendor-defined sidebands

Management Innovations - CID and SCD

CID — CONNECTOR INTERFACE DEFINITION

Traditionally: mechanical keying. Physical keys decide what may plug in where — rigid and fixed at tooling time. Not scalable. **> 50+ Potential Configurations.**

With OAC: the adapter declares itself. A structured document describing connector population, protocol, lane map, clocking and power.

- Interface and protocol flexibility
- Incompatibility caught in software, not by a key
- A profile collapses a full pin map to one ID

SCD — SLOT CONFIGURATION DATA

Traditionally: strap pins. Configuration arrives as a handful of resistor-strapped pins, and every bit carries a cost. Example: OAM has **13 straps**.

With OAC: the platform declares the slot. Slot data is delivered on that same path. Bits become kilobytes.

- Configuration flexibility with no pins spent
- New attributes cost a schema rev, not a respin
- One adapter adapts to whichever slot it lands in

THE EXCHANGE, SIMPLIFIED

1 OAC_MC_RDY
The adapter's controller is alive on AUX power

›

2 CID retrieved
Platform reads the adapter's declaration over PLDM Type 4

›

3 Compatibility
Decided in software, against this slot

›

4 SCD written
Slot context delivered — before primary power

›

5 OAC_PWR_EN
Validation passed; only now does main power flow

Extensibility, at lower cost and complexity: Every pin spent on configuration costs money, adds routing and test complexity. CID and SCD move that exchange into the management plane — so configuration becomes more extensible without changes to connector pinouts.

OCP Hardware Management Alignment

OCP GPU & Accelerator Hardware Management Interfaces	OCP GPU & Accelerator RAS Requirements	OCP GPU Firmware Update Specification	OCP Open Boot and Management Framework (OBMF-ICP over USB)
The management model itself. An accelerator-class OAC presents the interfaces a platform already knows how to drive — sensors, thresholds, inventory and control, in the form OCP already defines.	**Errors on the same path.** Reporting, logging and recovery ride the one USB link.	**One update mechanism.** The same interface carries firmware update across every compliant adapter, whoever built it.	**Binding over the same wire.** OBMF-ICP consolidates boot and management interfaces over USB 2.0.

OAC is grounded in these OCP specifications. Existing OCP management solutions will carry over to OAC platforms with little or no change.

Key Takeaways

1. **Designed from a blank slate.** Nothing carried forward because it was already there — legacy adapter-management practice was deliberately left behind.

2. **One mandatory management path.** MCTP over USB 2.0 with PLDM on CONN0 Columns 10–11 — one transport, always present, always fully capable.

3. **Identity before power.** CID is read on auxiliary power and validated before OAC_PWR_EN is ever asserted.

4. **Configuration is data, not straps.** SCD turns a handful of strapped bits into kilobytes — new attributes cost a schema revision, not a respin.

5. **OCP grounded.** OAC implements the OCP GPU & Accelerator Management, RAS, and Firmware Update specifications.

Call To Action

- Support for open, modular platforms continues to drive innovation and new ideas. Get involved, provide feedback, help to grow an OAC ecosystem.

- Specification Timeline
  - v0.9 - Available Today
  - https://www.opencompute.org/w/index.php?title=Server/OpenAdapterCard

- Provide feedback
  - AMD – Ross Hamilton ross.hamilton@amd.com
  - AMD – Justin King justin.king@amd.com
  - Intel – Marko Bartscherer marko.bartscherer@intel.com




Thank
You!
OCP
GLOBAL
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October 12-15, 2026
San Jose, California